Example: Three Related Computer Science Thesis Ideas

General Area: *AI-Based Flood Disaster Management and Decision Support*

Thesis Idea 1 — Street-Level Flood Prediction

Area of Interest: Artificial Intelligence, Machine Learning, Spatiotemporal Data Analytics

Real-World Problem: Flood warnings are often issued at the city or barangay level, but actual flooding can vary significantly from one street to another. Residents and disaster-response personnel need more localized and timely predictions.

Target Users / Environment: LGUs, disaster risk reduction offices, emergency responders, motorists, and residents in flood-prone communities.

Available Data: Historical rainfall, flood records, elevation/DEM data, drainage information, river/water-level data, weather forecasts, road-network data, and geographic coordinates of previous flooding incidents.

Computational Task: Predict the **probability, depth, and expected severity of flooding for specific road segments** within the next 30–120 minutes.

Possible Method: Spatiotemporal Graph Neural Network (ST-GNN) combined with LSTM/Transformer-based time-series modeling. Road segments can be represented as nodes in a graph, with neighboring roads, waterways, elevation, and drainage connections represented as edges.

Evaluation Metrics: MAE/RMSE for predicted flood depth; Precision, Recall, F1-score, and AUC for flood/no-flood classification; prediction lead time; comparison against LSTM, Random Forest, and XGBoost baselines.

Possible Research Contribution: A localized flood-prediction model that considers both **temporal weather behavior and spatial relationships among roads, waterways, elevation, and drainage systems** rather than predicting flooding only at a general geographic level.

Working Thesis Title:
"A Spatiotemporal Graph Neural Network for Street-Level Flood Depth and Risk Prediction Using Multisource Geospatial and Meteorological Data"

Related Published Studies

1. Yuan, F., Xu, Y., Li, Q., & Mostafavi, A. (2022). *Spatio-temporal graph convolutional networks for road network inundation status prediction during urban flooding*. **Computers, Environment and Urban Systems, 97**, 101870. DOI: 10.1016/j.compenvurbsys.2022.101870.
    - Used STGCN models to predict near-future inundation status at the **individual road-segment level**, making it highly relevant to the proposed topic.
2. Kazadi, A., Doss-Gollin, J., Sebastian, A., & Silva, A. (2024). *FloodGNN-GRU: A spatio-temporal graph neural network for flood prediction.* **Environmental Data Science, 3**, e21. DOI: 10.1017/eds.2024.19.
    - Combined graph neural networks with GRU modeling to capture both spatial and temporal flood dynamics.
3. Zhou, Y., Wu, W., Nathan, R., & Wang, Q. J. (2021). *A rapid flood inundation modelling framework using deep learning with spatial reduction and reconstruction.* **Environmental Modelling & Software, 143**, 105112. DOI: 10.1016/j.envsoft.2021.105112.
    - Demonstrated the use of deep learning and LSTM-based modeling for rapid prediction of flood-inundation time series.
4. Yang, F., Ding, W., Zhao, J., Song, L., Yang, D., & Li, X. (2024). *Rapid urban flood inundation forecasting using a physics-informed deep learning approach.* **Journal of Hydrology, 643**, 131998. DOI: 10.1016/j.jhydrol.2024.131998.
    - Integrated hydrodynamic simulation and deep learning to provide rapid, grid-based urban flood-depth forecasting.
5. Haces-Garcia, F., Ross, N., Glennie, C. L., Rifai, H. S., Hoskere, V., & Ekhtari, N. (2025). *Rapid 2D hydrodynamic flood modeling using deep learning surrogates.* **Journal of Hydrology, 651**, 132561. DOI: 10.1016/j.jhydrol.2024.132561.
    - Evaluated several deep neural-network architectures as faster surrogate models for high-resolution flood-depth simulation.

Possible Gap Identified from the Studies:

Existing studies demonstrate the feasibility of graph neural networks and deep-learning surrogate models for flood prediction. A possible research direction is to investigate whether **road topology, elevation, drainage connectivity, rainfall, water-level observations, and other real-time data can be fused into one road-level spatiotemporal graph model** for shorter-term and highly localized prediction.

Thesis Idea 2 — Automatic Road Passability Detection During Flooding

Area of Interest: Computer Vision, Deep Learning, Intelligent Transportation Systems

Real-World Problem: During floods, motorists and emergency responders may not know whether a road is safe to cross. Existing maps may still indicate that roads are accessible even when floodwater has already made them dangerous or impassable.

Target Users / Environment: Emergency responders, LGUs, motorists, ambulance operators, rescue teams, and traffic-management personnel.

Available Data: CCTV footage, road-camera images, drone footage, crowdsourced images, rainfall records, road elevation, vehicle information, and annotated images showing different flood levels.

Computational Task: Detect flooded roads, estimate approximate floodwater depth, and classify roads as **Passable, High Risk, or Impassable**.

Possible Method: YOLO/RT-DETR for vehicle and reference-object detection combined with monocular depth estimation or semantic segmentation. A classification model can then determine road passability using estimated water depth, vehicle type, and road characteristics.

Evaluation Metrics: mAP for object detection; IoU/Dice Score for water segmentation; MAE for estimated water depth; Accuracy, Precision, Recall, and F1-score for road-passability classification; inference speed/FPS.

Possible Research Contribution: A computer-vision framework capable of transforming ordinary CCTV or roadside-camera images into **real-time road-passability information without requiring dedicated flood sensors at every location**.

Working Thesis Title:
"Vision-Based Flood Depth Estimation and Dynamic Road Passability Classification Using Deep Object Detection and Monocular Depth Analysis"

Related Published Studies

1. Alizadeh Kharazi, B., & Behzadan, A. H. (2021). *Flood depth mapping in street photos with image processing and deep neural networks.* **Computers, Environment and Urban Systems, 88**, 101628. DOI: 10.1016/j.compenvurbsys.2021.101628.
    - Used deep learning and image processing to estimate floodwater depth from street-level photographs using submerged stop signs as reference objects.
2. Lopez-Fuentes, L., Farasin, A., Zaffaroni, M., Skinnemoen, H., & Garza, P. (2020). *Deep learning models for road passability detection during flood events using social media data.* **Applied Sciences, 10**(24), 8783. DOI: 10.3390/app10248783.
    - Directly investigated deep-learning classification of flooded roads according to whether they were passable or not.
3. Ahmad, K., Pogorelov, K., Riegler, M., Ostroukhova, O., Halvorsen, P., Conci, N., & Dahyot, R. (2019). *Automatic detection of passable roads after floods in remote sensed and social media data.* **Signal Processing: Image Communication, 74**, 110–118. DOI: 10.1016/j.image.2019.02.002.
    - Used deep models, CNNs, satellite imagery, and social-media imagery to identify passable and non-passable roads after floods.
4. Park, S., Baek, F., Sohn, J., & Kim, H. (2021). *Computer vision-based estimation of flood depth in flooded-vehicle images.* **Journal of Computing in Civil Engineering, 35**(2), 04020072. DOI: 10.1061/(ASCE)CP.1943-5487.0000956.
    - Proposed a computer-vision approach that used flooded vehicles, Mask R-CNN, and image retrieval to estimate water depth.
5. Song, Z., & Tuo, Y. (2021). *Automated flood depth estimates from online traffic sign images: Explorations of a convolutional neural network-based method.* **Sensors, 21**(16), 5614. DOI: 10.3390/s21165614.
    - Developed an automated framework using Mask R-CNN and traffic signs for estimating flood depth from images.

Possible Gap Identified from the Studies:

Prior studies separately demonstrate **flood-depth estimation, object/reference detection, and road-passability classification**. A possible gap to investigate is an integrated model that combines **real-time CCTV analysis, water segmentation/depth estimation, vehicle characteristics, and dynamic passability classification** within a single computational framework.

Thesis Idea 3 — Intelligent Dynamic Evacuation Routing

Area of Interest: Artificial Intelligence, Graph Algorithms, Reinforcement Learning, Decision-Support Systems

Real-World Problem: Conventional navigation systems usually identify the shortest or fastest route but may fail during disasters because road conditions change rapidly. A previously safe evacuation route may become flooded while evacuees are already traveling.

Target Users / Environment: Disaster-response agencies, evacuation coordinators, emergency vehicles, rescue teams, and residents of flood-prone communities.

Available Data: Road networks, evacuation-center locations and capacities, predicted flood depth, rainfall, traffic conditions, elevation, historical flood records, road-passability information, and travel-time data.

Computational Task: Continuously determine the **safest evacuation route** while considering changing flood conditions, traffic congestion, travel time, road accessibility, and evacuation-center capacity.

Possible Method: Model the road network as a dynamic weighted graph. Combine **multi-objective route optimization with Deep Reinforcement Learning**, where road-edge costs change according to predicted flood risk, congestion, travel time, and road passability.

Evaluation Metrics: Evacuation success rate; average evacuation time; percentage of routes exposed to high-risk flooded roads; computational response time; route stability; cumulative risk score; comparison with Dijkstra, A*, and conventional shortest-path algorithms.

Possible Research Contribution: A dynamic routing algorithm that does not simply identify the shortest path but **learns to balance travel time, flood exposure, congestion, road accessibility, and evacuation-center capacity under continuously changing disaster conditions**.

Working Thesis Title:
"Risk-Aware Dynamic Evacuation Route Optimization Using Deep Reinforcement Learning and Spatiotemporal Flood Predictions"

Related Published Studies

1. He, M., Chen, C., Zheng, F., Chen, Q., Zhang, J., Yan, H., & Lin, Y. (2021). *An efficient dynamic route optimization for urban flooding evacuation based on Cellular Automata.* **Computers, Environment and Urban Systems, 87**, 101622. DOI: 10.1016/j.compenvurbsys.2021.101622.
    - Developed a Cellular Automata-based Dynamic Route Optimization algorithm that updates evacuation routes according to evolving flood conditions.
2. Li, X., Liang, X., Wang, X., Wang, R., Shu, L., & Xu, W. (2023). *Deep reinforcement learning for optimal rescue path planning in uncertain and complex urban pluvial flood scenarios.* **Applied Soft Computing, 144**, 110543. DOI: 10.1016/j.asoc.2023.110543.
    - Applied deep reinforcement learning to flood rescue path planning while incorporating uncertainty, safety, and timeliness.
3. Li, X.-Y., & Wang, X. (2025). *Rescue path planning for urban flood: A deep reinforcement learning-based approach.* **Risk Analysis, 45**(4), 928–943. DOI: 10.1111/risa.17599.
    - Proposed an enhanced reinforcement-learning approach designed to handle uncertain flood risks and complex obstacles during rescue path planning.
4. Yang, W., Yan, X., Hu, D., Jiang, X., Song, J., Lang, R., Xu, K., & Lian, J. (2025). *A novel emergency evacuation route optimization model in flood disasters using hydrodynamic model and intelligent algorithm.* **Safety Science, 182**, 106709. DOI: 10.1016/j.ssci.2024.106709.
    - Coupled flood hydrodynamic modeling with ant-colony and genetic-algorithm approaches while accounting for congestion in evacuation routing.
5. Liang, J., Liu, M., Zhang, Z., & Yang, L. (2025). *Shortest path planning and dynamic rescue forces dispatching for urban flood disasters.* **Scientific Reports, 15**, 23643. DOI: 10.1038/s41598-025-06374-2.
    - Combined a customized A* path-planning algorithm with dynamic allocation of rescue resources based on real-time flood conditions.

Possible Gap Identified from the Studies:

Existing research already covers dynamic routing, Cellular Automata, intelligent optimization, A*, and Deep Reinforcement Learning. Therefore, simply proposing a **"Flood Evacuation Routing System" would not be enough**. A stronger research direction would investigate a model that integrates **predicted future flood conditions,**

real-time road passability, congestion, travel time, and evacuation-center capacity into a continuously updating multi-objective reinforcement-learning framework.

Strongest Idea: Thesis Idea 3

Why is this problem worthy of a Computer Science thesis?

The problem requires more than developing a conventional mapping or navigation application because evacuation routing must respond computationally to **dynamic and uncertain environmental conditions**. It involves graph optimization, artificial intelligence, real-time data processing, and multi-objective decision making to determine routes that balance safety, distance, congestion, and flood risk. The proposed research can also experimentally compare the developed algorithm against established routing techniques such as Dijkstra and A*. Therefore, the main contribution is not merely the resulting application, but a **new or improved computational approach for risk-aware dynamic routing during flood emergencies**.